

Proposal — ExpertAttention: Output-Conditioned Aggregation for Mixture-of-Experts Transformers

Abhay Sashta S | +91 99286 71118 | sabhaysashta@gmail.com | [in](#) [LinkedIn](#)

The Idea

Standard sparse Mixture-of-Experts (MoE) transformers compute aggregation weights from the input *before* any expert runs: $\text{output} = \sum_i g_i(x) \cdot \text{expert}_i(x)$. The router cannot see what experts produce — it predicts relevance from the input alone, and those same predictions determine the final combination. This conflates two distinct decisions: *which experts to activate* and *how to combine their outputs*.

We propose **ExpertAttention (EA)**: a lightweight attention mechanism that replaces the fixed weighted sum with output-conditioned aggregation. Routing is unchanged. After the top-k experts run, EA computes $Q = W_q \cdot x$, $K = \text{RMSNorm}(W_k \cdot E)$, $V = W_v \cdot E$ over actual expert outputs E , and produces aggregation weights $\alpha = \text{softmax}(QK^T / \sqrt{d_{\text{head}}})$. Compute overhead is <0.003%. Weights are recomputed every forward pass — at inference, the same input routed to the same experts can receive different aggregation weights depending on what those experts actually produced for that token.

The Hypothesis

Output-conditioned aggregation breaks the dominance feedback loop documented in MoE Lens (top-1 expert $\approx$ full ensemble in trained MoE), produces more complementary expert outputs, and yields lower perplexity than standard MoE — especially as expert specialisation increases with k and scale.

Current Results (84M params, k=2, 30K steps, C4)

- Cold-start failure observed.** Introducing EA from step 0 *prevents* expert specialisation entirely. Router entropy H_r stayed frozen at 0.692 (maximum) for the entire run, vs the baseline's healthy $0.637 \rightarrow 0.501$ curve. Validation PPL gap widened monotonically from +30% at step 5K to +68% at step 30K (503 vs 299). Root cause: EA's uniform aggregation at initialisation creates a gradient shortcut that removes the asymmetry needed to start the dominance loop. The underlying *dynamic* (uniform routing $\leftrightarrow$ homogeneous experts as a self-reinforcing loop) is documented in the broader MoE specialisation-collapse literature. The specific *trigger* — that introducing a learnable output-conditioned aggregation mechanism at step 0 produces this collapse in standard top-k MoE — has not, to our knowledge, been previously characterised.
- Scheduled residual ramp fixes it.** Output: $\lambda \cdot \text{standard} + (1-\lambda) \cdot \text{EA}$, $\lambda=1$ during freeze (steps 0–10K), then λ decaying $1 \rightarrow 0$ over 8K steps. **Freeze and ramp timing were derived from the baseline's H_r curve, not tuned blindly.** The Run 1 baseline showed a sharp specialisation transition at step 9K (val PPL dropped 21% in one checkpoint with no corresponding CE change — the moment expert specialisation began to translate into generalisation), with H_r approaching its plateau (~ 0.51) by step 18K. We chose freeze-end = 10K (just past the specialisation onset) and ramp-end = 18K (when specialisation saturates) so EA only takes over after experts have differentiated. We tried both **linear** and **cosine** ramp shapes. At step 26K (the latest checkpoint reached by all variants), val PPL was: linear 275.89, cosine 275.89, hard-switch freeze 323.76, baseline 301.87. The two ramp shapes are essentially identical — *the existence of a ramp matters, the shape does not*. Both ramp variants beat the hard-switch freeze and the baseline.
- EA learns an independent signal.** EA-router correlation traced an arc of $-0.38 \rightarrow 0.0 \rightarrow +0.52$: EA initially *disagrees* with the router, then refines to selective agreement. This is a learned policy the input-only router cannot express.
- At small scale, EA acts as an informed re-ranker.** Effective expert count ≈ 1.45 (max 2.0), dominance ratio 0.81. With expert output cosine similarity at 0.47, there is little to combine — EA rationally adapts by becoming a sharper selector than the router can be.

What Needs Work

The 84M-parameter results validate the *mechanism*: that output-conditioned aggregation is trainable, that the cold-start failure exists and is fixable, and that EA learns a signal independent of the router. Mechanism validation at this scale is the right scope — downstream task benchmarks (HellaSwag, ARC, MMLU, etc.) are near-random below $\sim 1\text{B}$ active parameters and would be uninformative here. The downstream-quality story requires scaling up; the mechanism story does not.

- Dynamic-weighting analysis** — live `disagree_rate`, `disagree_win_rate` (does EA's override beat the router?), `weight_shift`, `adapt_corr` (does EA respond to per-token output diversity?). Implemented.
- Scale-up to $\geq 1\text{B}$ active params on SlimPajama with downstream evals** (HellaSwag, ARC, PIQA, WinoGrande, MMLU, per-domain perplexity) — required to demonstrate that the mechanism translates into measurable downstream quality gains.

The EDL Argument (and Why We Have Not Run It Yet)

SD-MoE (arXiv:2602.12556) and Expert Divergence Learning (EDL, arXiv:2603.00054) establish that experts in standard MoE training share dominant spectral subspaces and learn redundant functionalities — a fundamental property of the training dynamics that persists at 7–15B parameters. EA's value depends on experts producing genuinely different outputs to condition on. **EDL should be combined with EA:** EDL forces expert diversity via a JSD auxiliary loss over per-domain routing distributions; EA exploits that diversity at aggregation time. The hypothesis is that the combination is superadditive — neither alone solves both homogenization and blind aggregation.

EDL has not yet been integrated due to compute constraints. All initial runs were performed on Kaggle T4×2 (free tier, 12-hour wall clock), forcing validation on smaller models (84M active) and a single-domain dataset (C4) where the per-domain labels EDL needs are unavailable. The cold-start failure and the ramp fix are real findings under these constraints; the next phase of the work requires GPU resources beyond what was available for the initial validation — the EDL+EA factorial at ≥ 1 B active parameters needs roughly 4× A100 80GB for ~2 weeks of training.

Selected Related Work

Aggregation is the unmodified default. Routing improvements (Expert Choice, Ada-K, Self-Routing, ReMoE, GRIN — arXiv:2202.09368, 2403.07652, 2604.00421, 2409.12136) all modify *which* experts activate; none modify *how* their outputs are combined. Soft-MoE (Puigcerver et al., 2023, arXiv:2308.00951) mixes inputs before experts but breaks the KV cache. Geometry-Preserving Aggregation (SBA, arXiv:2602.14039) keeps router weights but uses a better combination formula; ExpertAttention is orthogonal — it changes the weights themselves. Quadratic Gating (arXiv:2410.11222) establishes a theoretical hierarchy where output-conditioned gating is strictly more expressive than any polynomial of x alone.

Routers are demonstrably miscalibrated. Routing Manifold Alignment (RoMA, arXiv:2511.07419) measures 10–20% accuracy gaps between learned routers and optimal routing on downstream tasks; semantically similar inputs receive dramatically different routing weights. "A Closer Look into MoE" (arXiv:2406.18219) finds routers select experts by output norm rather than output quality. PASs-MoE (arXiv:2601.13020) documents misaligned co-drift between router and expert quality during training. The Illusion of Specialization (arXiv:2601.03425) identifies a "Standing Committee" of experts capturing the majority of routing mass across all domains.

Expert outputs converge under standard training. SD-MoE (arXiv:2602.12556) shows expert parameters share dominant spectral subspaces with similarity 0.7–0.9 across DeepSeek and Qwen at 7B; gradient subspaces are aligned because of low-rank structure in natural language data — scaling alone does not fix this. "A Closer Look" reports cosine similarity > 0.8 between selected expert outputs in late layers; "Diversifying the MoE Representation" (arXiv:2310.09762) reports $> 99\%$ similarity in some settings. Expert Divergence Learning (arXiv:2603.00054) addresses this with a domain-conditioned JSD auxiliary loss and shows consistent gains at 3B, 8B, and 15B scales.

Expert specialisation does emerge — just not in output magnitudes. The Expert Strikes Back (arXiv:2604.02178) confirms experts develop fine-grained task-level specialisations (morphological operations, syntactic continuation, semantic operations) that become more monosemantic under sparser routing. OLMoE (arXiv:2409.02060) at 7B/1B-active demonstrates clear domain and vocabulary specialisation with experts rarely co-activated. Specialisation lives in the routing patterns and operational roles, not the output magnitudes — which is precisely the gap output-conditioned aggregation is designed to exploit.

Output-conditioned signals demonstrably help when used. Dynamic routing (arXiv:2403.07652) activates more experts when router confidence is low, achieving improvements with fewer activated parameters. SERE (arXiv:2602.07616) re-routes tokens by exploiting output similarity. GatePro (arXiv:2510.13079) optimises selection for output diversity. Bidirectional routing (arXiv:2406.00023) feeds post-execution efficiency affinity back into routing. None apply output-conditioning to the aggregation step itself within LLM pretraining — ExpertAttention is, to our knowledge, the first.

Methodological grounding. MoE Lens (arXiv:2603.05806) establishes that in trained MoE the top-1 expert output approximates the full ensemble (cosine similarity ≈ 0.95) — the empirical signature of the dominance loop EA is designed to break. Attention Residuals (arXiv:2603.15031) demonstrates that RMSNorm-stabilised attention over upstream representations works at 48B/1.4T-token scale, validating the broader design principle EA applies on the expert axis. Gated Attention (Qwen, NeurIPS 2025 Best Paper) confirms that post-computation gating mechanisms are stable and scale.